A calibrated world model of first-name and last-name pairs, and its quantization

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Abstract. We model the probability that a person in country c has the first name f and the last name l, for 105 countries, 1.7 M first names and 3.5 M last names. The baseline is the independence model $p(f \mid c)\, p(l \mid c)$. We add a low-rank interaction between names with per-country gates, train it with an importance-sampled estimate of the partition function, and fit a scale and an offset per country on held-out people. The model is better than independence in all 105 countries, with a test gain of 0.467 bits per person. Product quantization with one byte for each pair of dimensions reduces the name vectors from 86 MB to 11 MB, for a loss of 0.0056 bits per person. All countries stay above the baseline.

1 Introduction

An anagram tool for person names must rank many candidate pairs (f, l) by how plausible they are in a given country. The independence model is a strong baseline, because most of the information is in the frequencies of the two names. The interaction between first and last names is weaker, but it exists: names from the same language or community occur together more frequently than independence predicts.

We have two requirements. (i) The model must not be worse than the baseline in any country: a good world average can hide losses in small countries. (ii) The model must be small enough for a web browser. Sections 3 to 5 give the model, its training and its calibration, which satisfy (i). Section 7 gives the quantization, which satisfies (ii).

2 Data and evaluation

The data is name_dataset [1], restricted to names in Latin script after NFC normalization and to names that occur 5 or more times in the world. It has 187 M unique triples (c, f, l) with a count of people n_{cfl}. For each triple we draw $n^{\text{test}} \sim \mathrm{Bin}(n_{cfl}, 0.1)$ and $n^{\text{valid}} \sim \mathrm{Bin}(n_{cfl} - n^{\text{test}}, 1/9)$, which gives 282 M train, 35 M validation and 35 M test people. The metric is the negative log-likelihood per person and the gain over the baseline p_0:

$$L = -\frac{1}{N} \sum_{i=1}^{N} \log_2 p(f_i, l_i \mid c_i), \qquad G = L_0 - L. \tag{1}$$

The baseline smooths the marginals of each country toward the world marginal π with α pseudo-people:

$$p_0(f, l \mid c) = \hat{p}(f \mid c)\, \hat{p}(l \mid c), \qquad \hat{p}(f \mid c) = \frac{N_{fc} + \alpha\, \pi_f}{N_c + \alpha}, \tag{2}$$

where N_{fc} counts the train people of country c with first name f (same for l). The value $\alpha = 10^5$ is the best on the validation split. The baseline has $L_0 = 25.68$ bits per person on the test split.

3 Model

The model adds an interaction to the baseline, only on the pairs whose two names occur in the train people of the country:

$$\log p(f, l \mid c) = \log p_0(f, l \mid c) + 1[(f, l) \in S_c] \left(\beta_c\, s_c(f, l) + b_c\right) - \log Z_c, \tag{3}$$

with the score $s_c(f, l) = \sum_{j=1}^{k} g_{cj}\, u_{fj}\, v_{lj}$. The vectors $u_f, v_l \in \mathbb{R}^k$, $k = 16$, are the same in all countries. Only names with $m = 20$ or more train people have a vector: 1.34 M of the 5.18 M names. The other names have $u = 0$. The gates $g_c \in (0, 1)^k$ let each country use a subset of the dimensions. The set $S_c = F_c \times L_c$ contains the pairs of names seen in country c; out of S_c the model is the baseline. The scale β_c and the offset b_c come from the calibration (Section 5); during training, $\beta_c = 1$ and $b_c = 0$. The partition function is

$$Z_c = 1 - P_c + \sum_{(f,l) \in S_c} p_0(f, l \mid c)\, e^{\beta_c s_c(f,l) + b_c}, \qquad P_c = p_0(S_c \mid c). \tag{4}$$

The sum has $\sum_c |F_c|\, |L_c| = 3.3 \times 10^{12}$ terms. We compute it exactly, in blocks, with MLX [2] on an Apple M4 Max GPU in about 3 minutes. All results in this report use the exact Z_c.

4 Training

The training loss is the negative log-likelihood of the train people (in nats). The baseline terms are constant, thus

$$\ell = -\frac{1}{N}\sum_i s_{c_i}(f_i, l_i) + \sum_c w_c \log Z_c, \tag{5}$$

where w_c is the fraction of people from country c. The sum in (4) is too large for each step. We write $Z_c = 1 - P_c + P_c\,\mathbb{E}_{p_0(\cdot\mid S_c)}[e^s]$ and estimate the expectation by self-normalized importance sampling [7], with $J = 4096$ samples per country and per step:

$$\begin{gathered}\mathbb{E}_{p_0(\cdot\mid S_c)}[e^s] \approx \frac{\sum_{j=1}^{J}\omega_j\, e^{s_j}}{\sum_{j=1}^{J}\omega_j}, \qquad \omega_j = \frac{p_0(f_j, l_j \mid S_c)}{q_c(f_j, l_j)}, \\ q_c = \tfrac{1}{3}\big(\hat{p}_{\text{train}}(\cdot \mid c) + p_0(\cdot \mid S_c) + t_c \otimes t_c\big).\end{gathered} \tag{6}$$

The first part of the proposal samples train people, where e^s is large. The second part is the target. The third part samples each name of country c in proportion to its number of distinct pairs, which gives more weight to rare names. Because $q_c \geq p_0(\cdot \mid S_c)/3$, all weights satisfy $\omega_j \leq 3$.

Two other choices failed. With sampled softmax (InfoNCE [3], 16 negatives), 19 countries were below the baseline, also on the train split: the interaction was too strong, and the cause was not overfitting. Without the threshold m, names with 3 to 9 people received large vectors under Adam, with scores up to 28 on pairs that no sample sees, and the estimate (6) was wrong. With $m = 20$, it agrees with the exact $\log Z_c$ within 0.002 in the countries that we checked. Settings: batches of 65,536 people, 10^4 steps of AdamW (learning rate 10^{-2}, weight decay 0.1), 2.7 minutes in total.

5 Calibration

For each country we fit (β_c, b_c) on the validation people. With $E = \mathbb{E}_{\text{valid}}[s\,1_S]$ and $Q = P_{\text{valid}}(S_c)$, the validation gain in nats is

$$G_c(\beta, b) = \beta E + b\,Q - \log\big(1 - P_c + e^b M_c(\beta)\big), \qquad M_c(\beta) = \sum_{(f,l)\in S_c} p_0(f, l \mid c)\, e^{\beta s_c(f,l)}. \tag{7}$$

For a fixed β, G_c is concave in b, with its maximum at

$$e^{b^\star} = \frac{Q\,(1 - P_c)}{(1 - Q)\,M_c(\beta)}. \tag{8}$$

One exact pass per country gives the histogram of s weighted by p_0 (bins of width 0.01). This gives $M_c(\beta)$ for all β in $\{0, 0.01, \ldots, 2\}$, and we keep the best pair, with b clipped to $[-3, 3]$. Because $(\beta, b) = (0, 0)$ is the baseline, the fitted validation gain is never negative. This is a per-country form of temperature scaling [4]. The median β_c is 0.86, and $\beta_c < 1$ in 81 of the 105 countries: the trained interaction is too strong in most countries.

6 Results

Table 1 shows the effect of each step. The calibration removes the last countries below the baseline and also increases the mean gain. Figure 1 shows the countries with the lowest gain before the calibration.

Model	Test gain (bits/person)	Countries below baseline
Shared vectors, sampled softmax	+0.303	20
+ country gates	+0.328	19
+ importance-sampled $\log Z_c$, $m = 20$	+0.409	4
+ calibration	**0.467**	**0**

Table 1. Test gain over the baseline (105 countries, 35 M people).

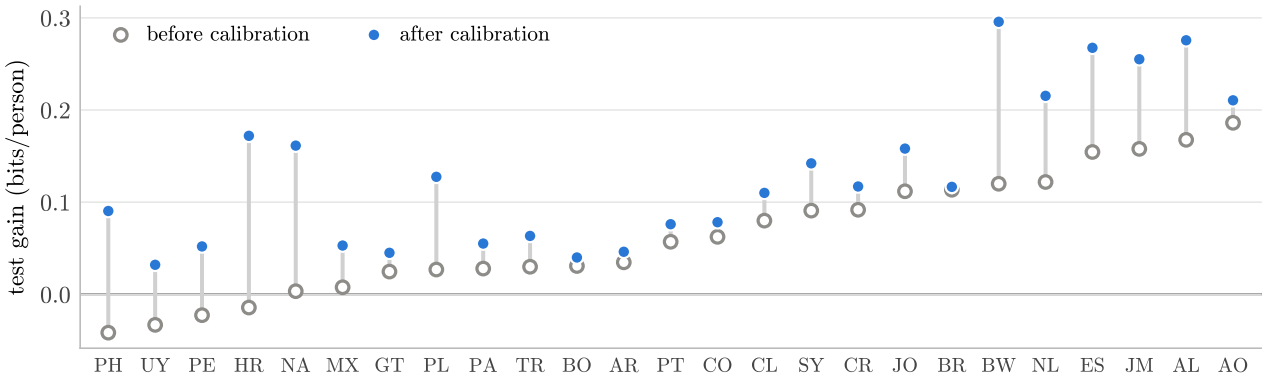


Figure 1. Test gain of the 25 countries with the lowest gain before the calibration (open circles) and after it (dots). The lowest country after the calibration is UY (+0.032).

7 Quantization

Only the name vectors need compression: U and V have 1.34 M rows of $k = 16$ numbers, 86 MB in float32. The gates and offsets are 105×17 numbers. A bit mask of the 5.18 M names (0.65 MB) tells which names have a vector.

7.1 Cost of an error on the vectors

Quantization changes the scores by $\delta = \hat{s} - s$. For one country, the change of the loss is exactly

$$\Delta L = -\,\mathbb{E}_{\text{data}}[\delta] + \log \mathbb{E}_p\left[e^{\delta}\right] = \big(\mathbb{E}_p[\delta] - \mathbb{E}_{\text{data}}[\delta]\big) + \tfrac{1}{2}\operatorname{Var}_p(\delta) + O(\delta^3), \tag{9}$$

where p is the model. Let $\varepsilon_f = \hat{u}_f - u_f$ and $\eta_l = \hat{v}_l - v_l$. To first order, the first term of (9) is $\sum_f \varepsilon_f^{\top} \nabla_{u_f} \ell + \sum_l \eta_l^{\top} \nabla_{v_l} \ell$, which is zero at a stationary point of the likelihood. If the errors of different names are independent with mean zero,

$$\Delta L \approx \tfrac{1}{2} \sum_f p(f)\, \varepsilon_f^{\top} A_f\, \varepsilon_f + \tfrac{1}{2} \sum_l p(l)\, \eta_l^{\top} B_l\, \eta_l, \qquad A_f = \mathbb{E}_p\big[(\beta g \odot v_l)(\beta g \odot v_l)^{\top} \,\big|\, f\big], \tag{10}$$

and B_l is the same with the roles of u and v exchanged. The cost of an error on a name is proportional to the frequency of the name. With $A_f \approx aI$, all the methods below minimize the error weighted by the count w_i of train people of each name i (row W_i of U or V):

$$D(\hat{W}) = \sum_i w_i \, \|\hat{W}_i - W_i\|^2. \tag{11}$$

Equation (10) is an approximation: the weight decay and the calibration move the model away from the stationary point.

7.2 Scalar quantization

Each dimension j has one scale. With b bits and $r = 2^{b-1} - 1$ [5],

$$\Delta_j = \frac{\gamma_j \max_i |W_{ij}|}{r}, \qquad q_{ij} = \operatorname{clip}\Big(\operatorname{round}\frac{W_{ij}}{\Delta_j},\, -r,\, r\Big), \qquad \hat{W}_{ij} = \Delta_j\, q_{ij}, \tag{12}$$

where the clip factor $\gamma_j \in \{0.4, 0.5, \dots, 1\}$ minimizes the term j of (11). Without clipping, the error is uniform on $[-\Delta_j/2, \Delta_j/2]$ with variance $\Delta_j^2/12$, thus by (10) $\Delta L \propto \Delta^2$. From 8 to 4 bits, Δ_j increases by about $127/7 \approx 18$, and (10) predicts a factor of about 330 on ΔL. We measure 4.8×10^{-5} and 0.010 bits per person: a factor of 209. The cost is $bk/8$ bytes per name.

7.3 Product quantization

Product quantization [6] cuts the k dimensions into k/g groups of g. Each group m has a codebook $C^{(m)} \in \mathbb{R}^{256 \times g}$, and each name stores one byte per group:

$$a_{im} = \arg\min_{a} \big\| W_i^{(m)} - C_a^{(m)} \big\|^2, \qquad \hat{W}_i = \big(C^{(1)}_{a_{i1}}, \dots, C^{(k/g)}_{a_{i,k/g}}\big). \tag{13}$$

The codebooks minimize (11) by weighted Lloyd iterations, from 256 rows drawn with probability proportional to w_i (25 iterations):

$$C_a^{(m)} \leftarrow \frac{\sum_{i:\, a_{im}=a} w_i \, W_i^{(m)}}{\sum_{i:\, a_{im}=a} w_i}. \tag{14}$$

The rate is $8/g$ bits per dimension. For $g = 2$ it is the same as 4-bit scalar quantization, but the 256 centroids follow the weighted joint density of the two dimensions, while the scalar grid puts its 15×15 cells on a square, many of them in empty regions. The loss is 0.0056 bits per person, against 0.010 for 4-bit scalar quantization. With $g = 4$ and $g = 8$ the loss is 0.049 and 0.120.

7.4 Truncation

The last method keeps the vectors of the n most frequent names only. The other names share the weighted mean $\bar{W} = \sum_i w_i W_i / \sum_i w_i$, which keeps their mean interaction. By (10), these names have a small $p(f)$, thus their error costs little. With 4-bit vectors for the top 50 % of the names (5.4 MB), the loss is 0.021, less than product quantization with $g = 4$ at the same size.

7.5 Choice

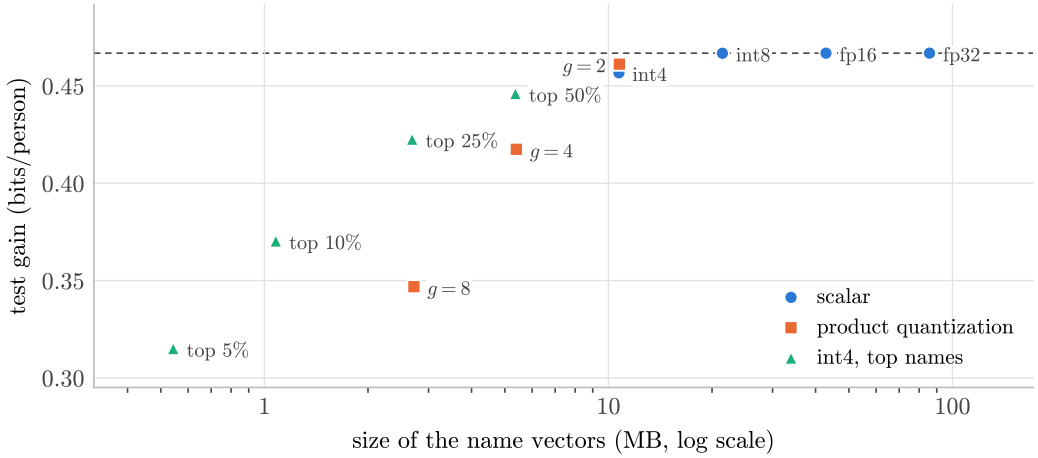


Figure 2. Test gain against the size of the name vectors. The dashed line is float32. Every variant keeps all 105 countries above the baseline.

Figure 2 shows the trade-off. Float16 and int8 have no measurable loss. At 11 MB, product quantization with $g = 2$ is better than 4-bit scalar quantization. Below 6 MB, truncation is the best method. We choose product quantization with $g = 2$, the smallest variant with a loss of 0.01 bits per person or less. The file, with codebooks, bit masks, gates and offsets, has 11.4 MB. Its exact test gain is 0.461 bits per person, and no country is below the baseline. The smallest variant (0.5 MB) still has a gain of 0.315, more than the float32 model trained with sampled softmax.

8 Limitations

The sizes do not include the count tables of the baseline, which the browser also needs. We did not test product quantization together with truncation. Some model choices were made with the test results visible, thus a small selection bias is possible.

References

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